Practical 5: Melting

On successful completion of this practical, you should be able to:

- Recognise and describe the mineralogy and textures of mantle rocks
- Explain the concept of mantle fertility
- Explain mantle melting reactions using a ternary phase diagram
- Interpret the provenance and history of mantle rocks

You will also practice the following skills:

- Describe the mineralogy and texture of igneous thin sections and hand specimens
- Classify igneous rocks by selecting an appropriate petrological classification scheme

Practical Reference Handout

Some useful pages for this practical:

- Classification and nomenclature of ultramafic rocks (p 11)
- Guide to systematic description of igneous rocks (p 2-4)

Don't try to interpret the order of crystallization in mantle rocks!

The Earth's mantle is solid. Mantle rocks undergo partial melting to produce basaltic melt. Depleted mantle rocks are the residue (leftover) material after partial melts have been extracted. Fertile mantle rocks have not yet undergone significant melting. The last time the whole mantle was molten was ~4.4 Ga, and this early Hadean magma ocean had fully cooled and solidified by ~4.2 Ga. Over the past ~4 Ga the original crystallization textures have been completely overprinted by solid-state processes including plastic deformation, diffusion, recrystallization, and Ostwald ripening. This means that it is not useful or meaningful to attempt to determine an order of crystallization for a mantle rock. Instead, look for polygonal textures with ~120° triple junctions between grains. This texture forms because it minimises the surface free energy at grain boundaries, but it requires long timescales to form by solid-state diffusion and recrystallization.

Part 1 (75 minutes): Mantle rocks in hand specimen and thin section

There are four samples to look at. For each sample, you should make notes and sketches of the important textural and mineralogical features, and estimate the modal mineralogy. The questions below are to guide your observations and help you to think about how to interpret what you are seeing. Aim to spend about 15 minutes on each sample.

<u>After 75 minutes, move on to Part 2 – even if you haven't finished looking at all the samples.</u>

A. Samples from ophiolites

Ophiolites are slices of oceanic lithosphere that, instead of being recycled into the Earth's mantle at a subduction zone, are preserved by being thrust onto continental crust. At the base of a typical ophiolite sequence is mantle peridotite. This peridotite is the source of the basaltic magma that goes on to crystallize and form the oceanic crust. The peridotite at the base of ophiolite sections has therefore undergone partial melting.

NC1 *New Caledonia*
The two main primary minerals in this rock are olivine and orthopyroxene. The olivine is extensively altered to pale green serpentine. The orthopyroxene contains exsolution lamellae of clinopyroxene. There is also minor spinel, which has a deep red-brown colour in plane-polarised light. Classify the rock based on the relative proportions of the main silicate minerals. Is your classification consistent with the tectonic setting of this sample?

Harzburgite. Yes, this is consistent with the rock being part of an ophiolite sequence. The mantle peridotite at the base of ophiolites has undergone partial melting, and harzburgite represents the residue left behind after partial melt has been extracted from lherzolite.

The texture of this rock indicates that it has been deformed (tectonised). What evidence for deformation is there? Suggest a cause of the deformation. Where might the water have come from which caused the serpentinisation?

Evidence for deformation: (1) Bent opx, shaped like **augens** (this is actually a metamorphic term), which are wrapped around by serpentine. The augen texture is also visible in the hand specimen. (2) Olivine appears to have been stretched and broken into subgrains (this is not always easy to see because of the extensive serpentinization). The deformation probably occurred when the oceanic lithosphere was obducted onto continental crust.

The water that caused the serpentinization of olivine is most probably seawater, in which case the serpentinization occurred either before or during ophiolite emplacement onto the continental crust. Further alteration could have occurred after emplacement due to circulation of meteoric water, although this alone could not explain the full extent of serpentinization in this sample.

B. Mantle xenoliths in kimberlites

BT2 **or** 15565 *Kimberley, South Africa* (hand specimens for BT2)
There are four main silicate minerals in these samples. The hand specimens of BT2 show the different colours of these minerals very clearly. The clinopyroxene is chrome diopside, and it has a distinctive bottle green colour due to its high chromium content. Look at the slide by eye and try to pick out the four minerals, including one pink mineral and one green mineral. Both samples contain minor phlogopite, a light brown mica that looks very similar to biotite. Identify the four main silicate minerals, estimate their modal proportions, and classify the rock.

Olivine, clinopyroxene, orthopyroxene, garnet. This is a garnet lherzolite.

Could this be a sample of fertile mantle?

Yes. “Fertile” means that the rock has previously undergone no (or little) partial melting; therefore it could undergo partial melting to produce a basaltic liquid. The residue left behind after melt extraction is referred to as depleted mantle, and contains fewer minerals (usually just olivine and opx). Rocks BT2 and 15565 contain cpx, garnet, and some minor phlogopite; all of these minerals would be consumed if the rock were to melt.

S.115 (no hand specimen)

The clinopyroxene is chrome diopside, and it has a distinctive green colour due to its high chromium content. The near-opaque mineral is spinel. The rock contains some large crystals of phlogopite, which is a light brown mica (it is the Mg-rich, Fe-poor endmember of the phlogopite-biotite solid solution). The presence of phlogopite is good evidence that this is a sample of fertile mantle: phlogopite is rich in K_2O and H_2O, both of which would partition strongly into the liquid if the rock was melted. What are the other two main minerals? Classify the rock.

Olivine, opx. The rock is a phlogopite spinel lherzolite.

The bottle-green mineral is chrome diopside. Pure diopside is a clinopyroxene with composition $CaMgSi_2O_6$. Chrome diopside has the formula $Ca(Mg,Cr)Si_2O_6$ – in other words, some of the Mg has been substituted by Cr. Chrome diopside is common in mantle xenoliths.

What does the mineralogy of this rock tell you about its depth of origin relative to that of sample 15565/BT2?

The aluminous phase in S.115 is spinel. The aluminous phase in BT2/15565 is garnet. Garnet is stable at higher pressure than spinel, so S.115 must come from a lower pressure than BT2/15565. We can say that S.115 came from the spinel stability field, which is ~0.8-2.0 GPa. We would need additional geochemical information (e.g. to use in the Al-in-opx barometer) to arrive at a more precise pressure estimate.

C. Orogenic peridotites

O32B *Finero*

There are two hand specimens. The larger one clearly shows a dark, shiny silicate mineral that picks out a distinctive foliation. What is this mineral?

Phlogopite

In thin section, the olivine in this sample contains deformation lamellae giving rise to patchy extinction. The dark brown isotropic mineral is spinel. What are the other two main silicate minerals?

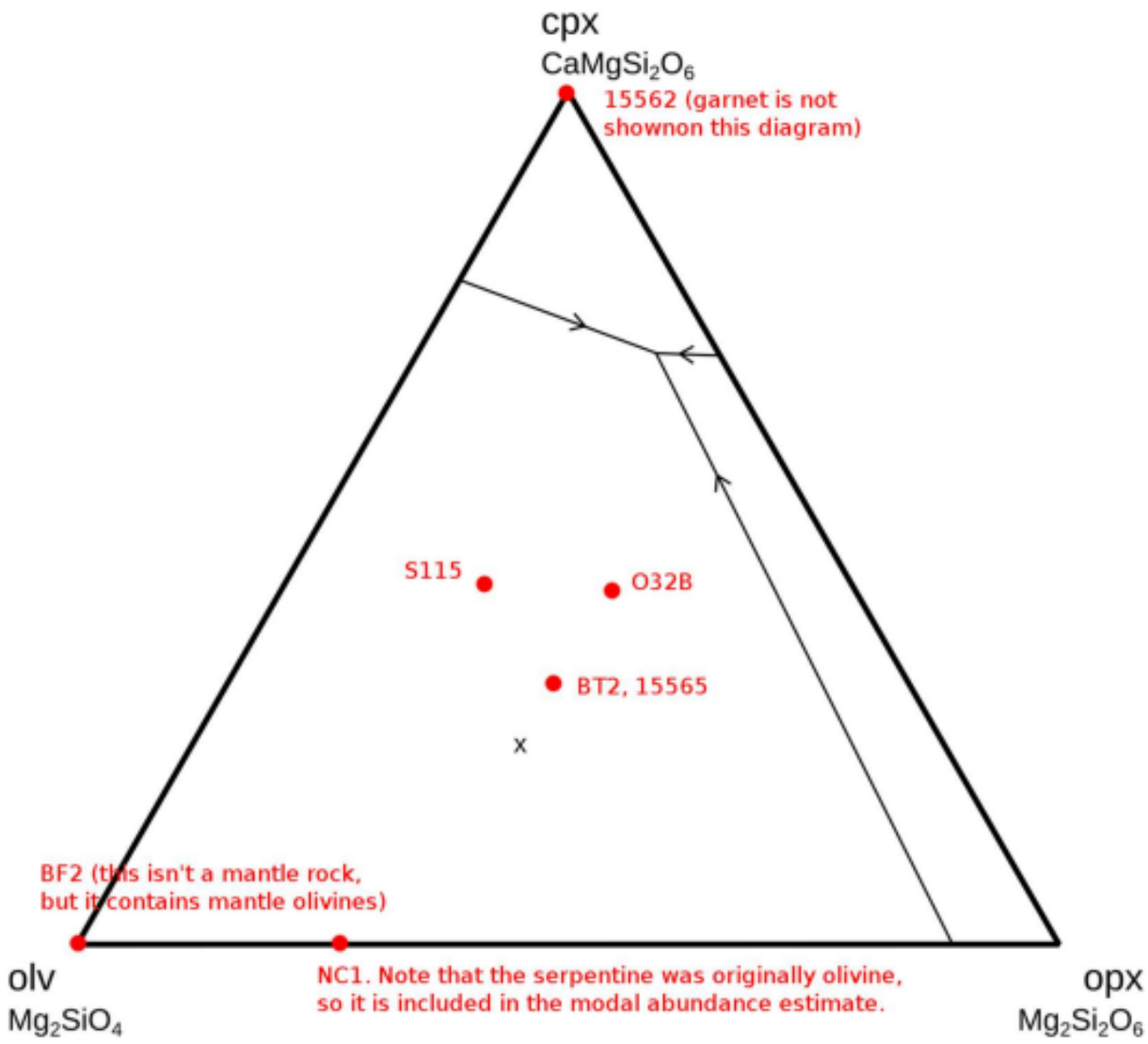
Clinopyroxene – pale green in PPL. Orthopyroxene – very pale brown-grey in PPL.

The dark brown isotropic mineral is spinel. There are two hand specimens; the larger one clearly shows the presence of another silicate mineral. What is it?

Part B (30 minutes): Mantle melting on a ternary phase diagram

You have now classified four different mantle rocks on the basis of their mineralogical and textural characteristics. Use your observed modal abundances to plot the four samples on a ternary phase diagram. Make sure each rock is labelled.

If you haven't finished Part 1, ask a demonstrator to help you plot any samples you haven't classified yet.



Your own plotted compositions will depend on your estimated modal abundances, and this will depend on which thin sections you looked at. However, your diagram should look broadly similar to this: NC1 on the bottom edge, and the other three samples somewhere in the middle of the triangle.

Check your answers with a demonstrator before you move on to the next set of questions.

1. Rock S.115 undergoes partial melting. Write down the melting reaction that will take place. What is the composition of the first melt to form? Express your answer in percentages of the three mineral components.

Melting will occur at the ternary eutectic. Olv + Cpx + Opx → L. The composition of the first melt is the same as the composition of the ternary eutectic: 6% olv, 69% cpx, 24% opx.

2. Rock 15565/BT2 undergoes partial melting. How will the composition of the solid residue will change as melt is extracted? Which mineral will be consumed first (i.e. which mineral will run out first), and what is the composition of the solid residue when this mineral has been exhausted?

The composition of the solid residue evolves directly away from the ternary eutectic as melt is extracted. Cpx will be consumed first. Once cpx has run out, the residue will contain only olivine and opx. Your predicted proportions of olivine and opx will depend the modal abundance you estimated for BT2/15565. You should find that the residue has a composition fairly similar to sample NC1.

3. According to your diagram, which of the fertile peridotite samples could partially melt to produce a solid residue most similar to the observed mineralogy of NC1?

Your answer will depend on your modal abundance estimates for NC1 and the fertile peridotites.

4. What are the advantages and weaknesses of using this simple ternary phase diagram to understand mantle melting reactions?

Strengths: Ternary phase diagrams are simple to use, and very powerful. This diagram does a very good job of explaining the observed mineralogy of depleted mantle rocks.

Weaknesses: This particular ternary diagram simplifies the true mineralogy of real mantle samples. It does not show the aluminous (garnet, spinel) or hydrous (phlogopite) phases, all of which would be involved in the melting reactions.

Bonus thin sections

If you've finished and would like to look at something a bit different, take a look at these more unusual examples of mantle rocks and minerals.

BF2 *Bultfontein, South Africa*
Most kimberlites are fragmental rocks. This is not a typical kimberlite, either mineralogically or texturally. It comes from a small sill associated with a true kimberlite.

There are three main minerals in this sample. Two of these are non-silicates. The first is perovskite ($CaTiO_3$), a brown, isotropic, square-shaped mineral. What is the second non-silicate mineral? What does it tell you about the chemical composition of the magma?

Calcite: it has high relief, and very high birefringence (pale pink/green; may look like a sandy brown). Calcite is a carbonate mineral, $CaCO_3$. The most likely source of the carbonate is CO_2 dissolved in the kimberlite magma. This tells us that kimberlite magmas are extremely volatile-rich. (Kimberlites also have very high H_2O contents, but this is not preserved in the mineral record.)

Describe the texture of this rock, and the shape of the olivine crystals. Many of these olivine crystals are **xenocrysts**: they are derived directly from the mantle, and were incorporated into the kimberlite magma as it drilled through the mantle and lithosphere on its way to the surface. The olivine crystals have reaction rims of serpentine. Where might the water have come from that caused the serpentinisation?

The overall texture is porphyritic. Some of the olivines are euhedral, some are subhedral, and some are rounded. The rounded shapes suggest that these crystals were not in chemical equilibrium with the kimberlite magma, and have been partially resorbed (re-dissolved). The serpentinization surrounding the olivines is caused by a chemical reaction between olivine and hydrous fluids. The fluids come from the kimberlite magma itself.

15562 *Kimberley, South Africa*

This rock is a xenolith from a kimberlite. The rock type isS **eclogite**, and it has the same bulk chemical composition as basalt. Eclogites are formed by high-pressure metamorphism of basalt. Eclogite nodules in kimberlites are believed to have come from subducted and metamorphosed basaltic oceanic crust. The hand specimen shows the shape of the nodule very clearly.

What are the two main minerals in this rock? Note the reaction rims around one of these minerals. The dark green, square-shaped, isotropic mineral is spinel.

Cpx, garnet. Notice that these are the two minerals that are consumed by partial melting of fertile lherzolite.

Plot BF2 and 15562 on your ternary diagram. How do their compositions compare with the other mantle rocks in this practical?

BF2 plots at the olivine apex, and 15562 plots at the clinopyroxene apex. These compositions appear to be quite different from the other mantle samples in this practical. On the peridotite classification diagram they look like monomineralic rocks (BF2 would be a dunite; 15562 would be a clinopyroxenite) – but this is because the ternary diagram does not capture information about the other minerals present in these rocks.